



Aishu R 251001007-2025-IT ▾

A2**Started on** Friday, 2 October 2026, 7:49 PM**State** Finished**Completed on** Friday, 2 October 2026, 7:51 PM**Time taken** 2 mins 4 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     unsigned long long dp[n + 1];
8     dp[0] = 1;
9
10    for (int i = 1; i <= n; i++) {
11        dp[i] = dp[i - 1];
12
13        if (i >= 3)
14            dp[i] += dp[i - 3];
15    }
16
17    printf("%llu", dp[n]);
18
19    return 0;
20 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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